Scaling Student Participation in Balanced Blended Space with an Automated Knowledge System

AUTHORS

Kazi Rahimu Islam

Junior, Computer Systems Technology

MENTOR

David B. Smith

Artistic Director, Center for Holistic Integration

01. Introduction

Balanced Blended Space (BBS) explores how communication and collaboration flow across physical, virtual, and conceptual realities, with ongoing work spanning the BBS Syntax, the Blended Reality Performance System (BRPS), and creative sub-projects such as Blended Shadow Puppet Theatre. A major barrier to scaling student participation is access. Core BBS materials are distributed across multiple web pages GitHub repositories, and evolving documents. This fragmentation slows onboarding, increases search time, and makes it easy for students to rely on outdated or incomplete information when trying to learn BBS concepts or identify projects to join. This project builds an automated living knowledge base to centralize approved BBS resources and support a student-facing assistant that answers questions with citations. Using n8n, we design agentic AI based workflows to continuously collect and refresh source content, clean and structure it, chunk text for retrieval, generate embeddings, and store vectors in a Supabase database. On top of this pipeline, a retrieval-augmented generation (RAG) layer routes user questions, retrieves relevant source passages, and produces concise, evidence-grounded responses aligned with official BBS definitions and project descriptions.

The expected outcome is a scalable system that reduces onboarding friction, improves the accuracy and consistency of student information access, and decreases repetitive mentor workload. Future directions include formal evaluation of answer accuracy, improved routing and chunking strategies, and feedback mechanisms that allow controlled updates while preserving strict source approval and versioning.

02. Objective

- Build one central place where students can find approved BBS information without digging through many pages and repos.
- Set up n8n workflows that automatically collect, clean, and update BBS materials so the knowledge base stays current.
- Create a student-facing assistant that answers questions by pulling relevant passages from the approved sources and showing citations.
- Make it easier for new students to understand what BBS is and quickly find projects they can join.
- Design the system so it can be tested and improved later, including accuracy checks, better routing, and controlled updates.

03. Methodology

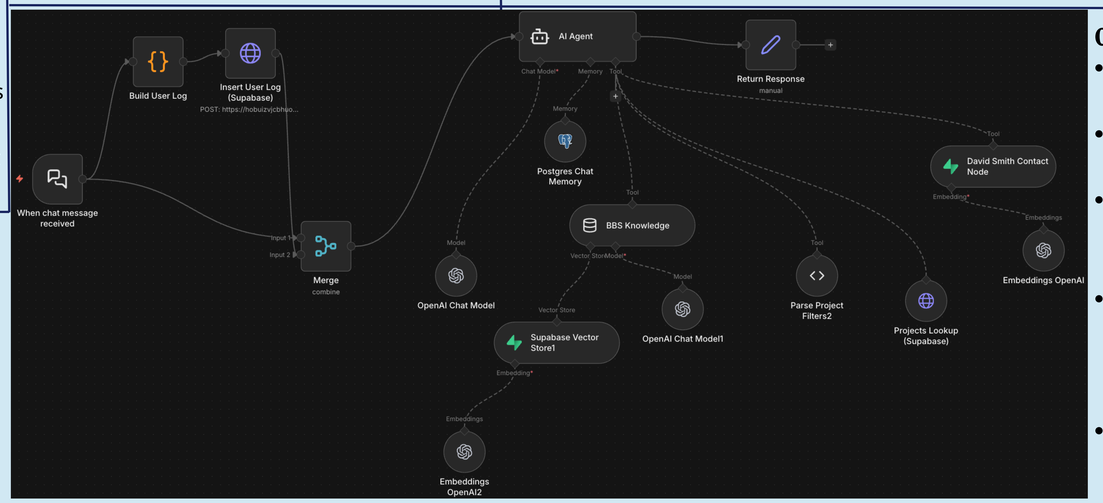
- Pick the only approved sources for the project, the BBS Google Sites pages and the CHI City Tech GitHub repositories, then list the exact pages and repos we want the system to cover.
- Build an n8n pipeline that pulls content on a schedule, scrapes or fetches text, and keeps a clean record of where each piece of content came from.
- Clean and standardize the text so it is usable for search, remove repeated sections, keep headings, and store useful metadata like source link, section title, and update time.
- Split the content into small, readable chunks that match how students ask questions, then generate embeddings for each chunk.
- Store chunks, metadata, and embeddings in a Supabase vector database so retrieval is fast and consistent.
- Add a RAG question answering layer that takes a student question, finds the most relevant chunks, and writes an answer only from those sources while showing citations.
- Add rules that prevent guessing, limit answers to what the retrieved sources say, and reject questions that are outside the approved materials.
- Test the workflow with a set of common student onboarding questions and track what sources were retrieved, whether the answer matches the source, and where the pipeline needs improvements.

04. Results/Findings

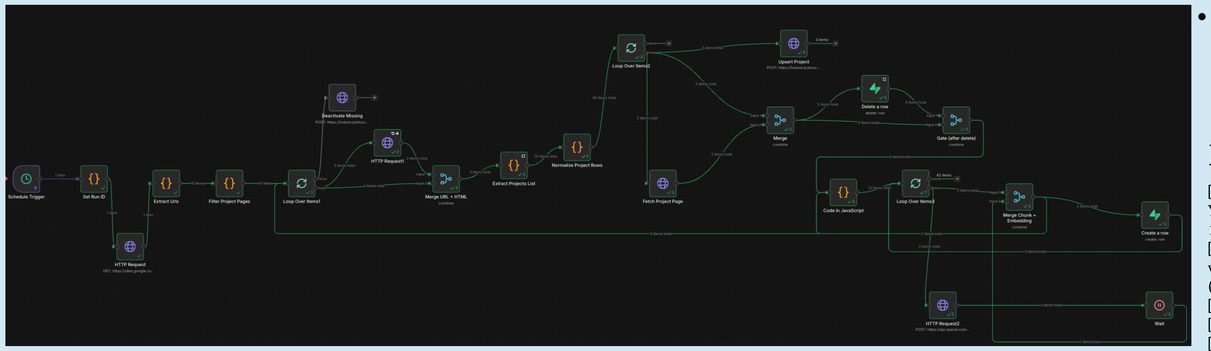
- Built an automated n8n pipeline that pulls approved BBS content from the BBS Google Sites and CHI City Tech GitHub.
- Cleaned and organized the content, then split it into searchable chunks for retrieval.
- Generated embeddings for the chunks and stored everything in a Supabase vector database for fast search.
- Connected a RAG question answering layer that retrieves relevant passages and answers with citations instead of guessing.
- Confirmed the full end to end workflow runs successfully and is ready for the next phase of student testing, accuracy checks, and improvements to routing, chunking, and answer rules.

05. Analysis

- We checked the full pipeline to see if it runs consistently.
- We confirmed the system is only pulling from the approved sources: the BBS Google Sites pages and the CHI-CityTech GitHub repos.
- We reviewed the cleaning step to make sure the text is readable and standardized.
- We tested chunking and retrieval by asking typical onboarding questions and making sure the top results actually match the question and come from the right page or repo.
- We audited the final answers to make sure the assistant isn't guessing, stays close to what was retrieved, shows citations, and highlights what still needs improvement (routing, chunk size, or missing coverage).



BBS Chatbot Agentic AI Agent



BBS Indexer Agentic AI Agent

06. Conclusion

- We turned BBS materials that were scattered and constantly changing into one reliable, searchable knowledge system.
- The pipeline automatically pulls only approved content from the BBS Google Sites and the CHI City Tech GitHub.
- It cleans everything up, organizes it, and stores it in a Supabase vector database so it can be searched quickly and intelligently.
- We also built a student facing assistant that answers questions by pulling exact passages from the sources and showing citations, so students can trust what they are reading.
- This connects to agentic AI because the goal is not just one time answers. It is a system that keeps knowledge fresh, routes questions correctly, and supports real workflows with less manual effort.
- In future, we will test it with real onboarding questions, measure accuracy and citation quality, and improve routing, chunking, and source coverage so it becomes a dependable tool for both students and mentors.

References

[1] K. Stratvert, "n8n Tutorial for Beginners: Build Your First Free AI Agent," YouTube video. [Online]. Available: https://youtu.be/dpoMEcXjVH8. [Accessed: Feb. 16, 2026]. (YouTube)

[2] "Step-by-Step: Build an n8n Website Chatbot That Reads Your Site," YouTube video. [Online]. Available: https://youtu.be/mLTqabG0l7c. [Accessed: Feb. 16, 2026]. (YouTube)

[3] n8n, "n8n Docs," n8n Documentation. [Online]. Available: https://docs.n8n.io/. [Accessed: Feb. 16, 2026]. (n8n Docs)

[4] Balanced Blended Space (BBS), "Home," Google Sites. [Online]. Available: https://sites.google.com/view/balancedblendedspace/home. [Accessed: Feb. 16, 2026]. (Google Sites)

[5] Center for Holistic Integration (CHI), "CHI-CityTech," GitHub organization page. [Online]. Available: https://github.com/CHI-CityTech. [Accessed: Feb. 16, 2026]. (GitHub)